

The pullback cancels the term exactly, yet cannot return to the physical metric for free

R0.70L left a natural question: can pulling back the vorticity covariance by the strain-generated matrix G completely remove the most dangerous stretching term? Algebraically, yes; but the stretching merely enters the evolving metric. Returning to ordinary Euclidean variables incurs a sharp $\kappa_2(G)^2$ factor. An affine-invariant metric removes that factor but requires positive-definite covariance, while the simplest smooth periodic shear already gives a rank-one boundary.

Status · R0.70M exact certificate, independent internal review PASS, and formal figure complete

Two direct pullback routes closed

Version v0.70M · 2026-08-24

Exact generators: all 4 groups passed

Figure validation: all 20 checks passed

Formal figure package: generic validation returned 0 errors / 0 warnings

Next gate: multiscale covariance frame coercivity

00 / EXACT CONCLUSION

Coordinate cancellation holds, but energy closure does not follow

Starting from the local filtered-vorticity covariance ledger

$$\dot{Q} = \Sigma Q + Q \Sigma + F, \quad \Sigma \in \text{Sym}_0(3),$$

Define the strain propagator and pulled-back covariance

$$\dot{G} = \Sigma G, \quad \hat{Q} = G^{-1} Q G^{-\top}, \quad \boxed{\dot{\hat{Q}} = G^{-1} F G^{-\top}}.$$

The constant-strain term does disappear completely. But this is only a change of coordinates; controlling the pulled-back residual from the NSE energy level and safely transferring the conclusion back to the original Euclidean metric remain two separate problems.

Conclusion at this stage. The pulled-back conditional estimate is exact; a universal return to the Euclidean metric nevertheless faces a sharp condition-number-squared obstruction. The affine inverse-covariance alternative, in turn, fails at a genuine rank defect of a smooth NSE solution.

01 / NAMING BOUNDARY

G is a strain propagator, not the physical deformation gradient

On this page,

$$\dot{G} = \Sigma G$$

It contains only the symmetric strain. I call it **strain propagator**, called the strain-only propagator in English. The deformation gradient of the physical flow map uses the full velocity gradient and satisfies

$$\dot{D} = (\Sigma + W)D.$$

In viscous Eulerian–Lagrangian formulations, diffusion commutators or stochastic averaging also appear. Thus, ‘pullback’ below refers only to the auxiliary matrix G , not to the Cauchy formula, and $\det G = 1$ is not interpreted as condition-number control.

02 / PULLED-SHAPE BV ESTIMATE

After removing pure amplitude, one obtains a conditional estimate with no additional Grönwall factor

Such that

$$\hat{E} = \text{tr } \hat{Q}, \quad \hat{B} = \frac{\hat{Q}}{\hat{E}} - \frac{1}{3}I.$$

Define the relative residual after removing the best scalar amplitude in the pulled-back metric

$$\rho_G = \inf_{\lambda \in \mathbb{R}} \frac{|G^{-1}(F - \lambda Q)G^{-\top}|_F}{\text{tr}(G^{-1}QG^{-\top})}.$$

The exact shape estimate is

$$|\widehat{B}(t) - \widehat{B}(s)|_F \leq (1 + \sqrt{2}) \int_s^t \rho_G(r) dr.$$

Thus $\rho_G \in L^1$ gives bounded variation and a finite terminal limit for $\widehat{B}$. What remains unavailable is a derivation of this integrability condition from Leray energy.

03 / SHARP EUCLIDEAN RETURN LOSS

The Euclidean-return factor $\kappa_2(G)^2$ cannot be reduced

If the same amplitude-optimized quotient ρ_0 is defined in the original coordinates, then

$$\rho_G \leq \kappa_2(G)^2 \rho_0.$$

An explicit positive-definite matrix family proves that, even after the pure-amplitude direction has been optimized out in both coordinate systems, one still has

$$\frac{\rho_G}{\rho_0} \longrightarrow \kappa_2(G)^2.$$

Symmetric strain gives only

$$\kappa_2(G(t)) \leq \exp \int (\lambda_{\max}(\Sigma) - \lambda_{\min}(\Sigma)) dt.$$

If this critical strain history is assumed finite at the outset, the argument returns to an existing strain-based continuation condition instead of deriving new control from the basic energy bound.

04 / ZERO-INTEGRAL HOLONOMY

A zero signed integral is not a zero history

Take two noncommuting symmetric trace-free matrices A, C , and apply four mutually disjoint smooth unit-mass pulses in the order $+A, +C, -A, -C$. The matrix integral is exactly zero, and the covariance residual is also set to zero, but the ordered exponential is

$$G_* = e^{-C} e^{-A} e^C e^A, \quad \text{tr } G_* = -\frac{862}{81} < -2.$$

It is hyperbolic and non-normal. The pulled-back covariance remains I , yet after one loop the physical three-dimensional covariance satisfies

$$\mathrm{tr}\, B_*^2 = \frac{2}{3} - \frac{13122}{3296483}.$$

A zero-integral, zero-residual loop already pushes the Euclidean shape close to the rank-one boundary. Time ordering and noncommutativity are responsible. At present this loop is only an exact smooth matrix history; it has not been embedded in an unforced finite-energy periodic NSE trajectory.

05 / AFFINE GEOMETRY AND RANK BOUNDARY

The affine-invariant metric removes the condition number, but requires positive-definite covariance

Let $\mathcal{G} = G^\top G$. Then

$$Q = G\widehat{Q}G^\top, \quad E = \mathcal{G} : \widehat{Q}, \quad \dot{\mathcal{G}} : \widehat{Q} = 2\Sigma : Q.$$

This shows that stretching has been transferred intact into the evolving metric. If $Q \succ 0$, standard affine-invariant SPD geometry can be used to define

$$\mathfrak{a}(Q, F)^2 = \mathrm{tr}[(Q^{-1}F)^2] - \frac{1}{3}[\mathrm{tr}(Q^{-1}F)]^2,$$

Its determinant-normalized shape velocity contains neither G nor $\kappa_2(G)$. This matrix geometry is established theory; at this stage I only place the exact NSE covariance residual in this metric and test the rank boundary.

Exact unforced periodic shear

$$u = A_0 e^{-\nu N^2 t} \sin(Ny) e_1$$

has vorticity that remains parallel to e_3 , so every nonzero covariance produced by scalar filtering has rank one. Q^{-1} already fails to exist for this simplest class of smooth NSE solutions; isotropic regularization also cannot pass uniformly to $\varepsilon \downarrow 0$.

06 / COMPLETE NSE RESIDUAL

All four classes of pulled-back residual terms remain at critical scale

Moving cutoffs, nonconstant strain, viscosity, and SGS commutators must all be retained. The direct estimate is

$$|\dot{\hat{B}}|_F \leq (1 + \sqrt{2}) (A_\chi + 2S_G + 2D_G + 2C_G).$$

The time integral of each term has Navier–Stokes critical scaling. On returning to Euclidean coordinates, the cutoff amplitude incurs no condition-number loss, nonconstant strain generally incurs a first-power loss, and viscosity and SGS generally incur a second-power loss. Leray energy currently controls none of these pulled-back critical quantities and gives no positive lower bound for $\hat{E}$.

07 / JOURNAL FIGURE

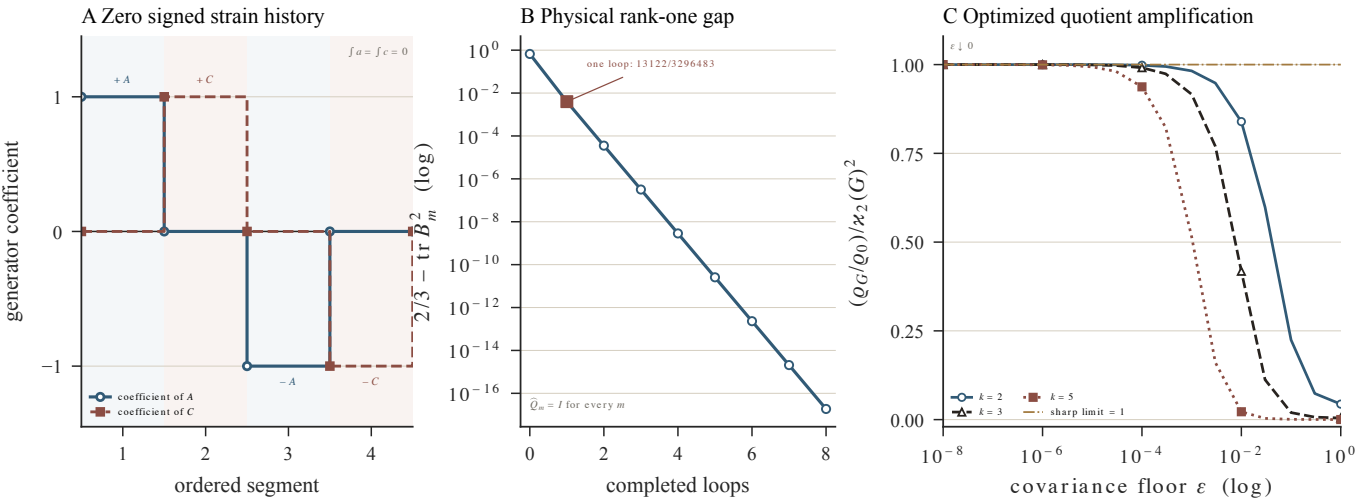
Noncommutative loop, rank-one gap, and sharp Euclidean amplification

Strain-only propagator holonomy and Euclidean residual amplification

zero signed history · fixed pulled covariance · sharp optimized kappa-squared loss



EXACT MATRIX ODE + PERIODIC-SHEAR RANK CHECK / HOLONOMY NOT YET AN UNFORCED FINITE-ENERGY NSE TRAJECTORY



pullback removes the source from Q-hat but stores stretching in the metric; affine-relative control still requires a coercive covariance frame

Figure Ro.70M. A: The generators are ordered $+A, +C, -A, -C$; both signed integrals vanish, but the ordered exponential gives an exact hyperbolic matrix. B: With $Q_0 = I$ and zero residual, the pulled-back covariance remains I , whereas the physical covariance approaches the rank-one boundary. C: For an explicit positive-definite family, the optimized residual quotient satisfies $\rho_G/\rho_0 \rightarrow \kappa_2(G)^2$. All curves come from closed-form matrix formulas, not DNS.

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08 / RESEARCH VALUE AND ROUTE DECISION

The value lies in splitting “the pullback might resolve stretching” into three verifiable facts

The coordinate cancellation is exact; the $\kappa_2(G)^2$ loss on returning to Euclidean coordinates is sharp; inverse-covariance geometry that removes the condition number then meets a genuine rank defect in a smooth periodic NSE flow. The value of Ro.70M is

therefore mainly route selection, not progress on regularity.

The core fluid-mechanics and PDE audit was limited to twelve high-signal primary sources, with three additional sources on affine SPD geometry. Physical deformation histories, critical continuation criteria, and matrix geometry all have clear precedents; the bounded search found no source containing this page's particular combination of local covariance, strain propagator, sharp quadratic loss, and rank-boundary conclusions. This is not a general novelty claim.

Ro.7oN will first test the minimal multiscale frame

$$\mathcal{Q}_k = \sum_{j=k-m}^{k+m} w_j Q_j, \quad \mathcal{Q}_k \succeq c \operatorname{tr}(\mathcal{Q}_k) I.$$

I will first derive its exact source and residual ledger, then use shear and Beltrami families to look for rank defects. Only after a coercive estimate passes these counterexample tests will I proceed to affine-relative estimates; no large-scale DNS will be launched before the algebraic gate is passed.

09 / CLAIM BOUNDARY

Exact matrix history and exact NSE boundaries must be kept separate

PROVED Pullback and conditional BV Source cancellation with no omitted terms and the optimized amplitude-quotient estimate.	PROVED sharp condition-number-squared loss and holonomy Both the positive-definite family and the smooth zero-integral matrix loop are certified exactly.
PROVED periodic NSE rank boundary Smooth unforced shear produces a rank-one covariance exactly.	OPEN Energy closure and trajectory realization The unforced trajectory realizing the loop, the weak-solution limit, and critical control remain incomplete.

This version does not claim that the four-pulse loop has been realized by an unforced finite-energy periodic NSE trajectory, does not claim that Leray energy controls the pulled-back residual, and does not obtain a new continuation criterion.

It proves neither blow-up nor global smoothness, and it does not solve the Millennium problem.

10 / REPRODUCIBILITY AND SOURCES

The exact generator, independent review, literature boundary, and formal figure package have all been archived

[View the exact certificate and SHA-256 manifest](#) · [View the formal journal-figure package](#) · [下载同步 PDF \(Chinese PDF\)](#) · [View the Clay Mathematics Institute's official problem statement](#)

The producer certifies the exact matrices, algebra, and rational arithmetic. The smooth realization of the matrix histories, the PDE identity for the periodic shear, the complete residual derivation, and the literature interpretation are separate analytic arguments; the formal manifest points to the immutable source snapshot and passes the generic figure-package validator.